

SANJANA KANCHIBOTLA

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Data Science, Machine Learning, Risk and Compliance Analytics

EXPERIENCE

CIBC

Jan 2026 – Jun 2026

ML Data Analyst, Intern | Toronto, ON

- Built a RAG-based exam-scoping assistant on AML compliance reports, reducing manual document retrieval from days to minutes for examiners operating under high-stakes regulatory timelines
- Mapped end-to-end examination workflows through two weeks of embedded research with the examiners, ensuring the system was designed around how they work
- Enabled natural-language search across multiple documentary repositories, surfacing ranked documents, prior issues, risk ratings, and emerging risk signals from a single query; roadmapped Databricks ingestion and a domain-news LLM module for proactive risk monitoring
- Presented to the VP and Senior Director of the Compliance and AML Monitoring team; defined a simulated evaluation framework for continuous quality improvement in the absence of a labelled baseline

PROJECTS

Credit Risk Prediction with Counterfactual Explanations

2025

Python, MLP, XGBoost, DiCE, Platt Scaling, SHAP, scikit-learn

[GitHub](#)

- Built a credit risk classifier on 148K mortgage applications achieving AUC-ROC of 0.901; conducted fairness auditing across gender, age and geographic subgroups
- Audited features for domain validity; identified and removed 5 variables whose predictive power stemmed from data artifacts rather than genuine borrower risk preventing the model from learning patterns that wouldn't hold in production
- Generated counterfactual explanations for declined applications showing the minimum most logical and feasible changes needed to flip a prediction, turning a black box model into an actionable decision tool
- Deployed on AWS SageMaker with a live monitoring layer that triggers retraining when feature drift exceeds 30%, making the model production-ready

Executive Compensation Anomaly Detection

2026

Python, SQL, Tableau, Compustat WRDS

[GitHub](#), [Tableau Dashboard](#)

- Detected executive compensation anomalies across 30 years of North American public company data, surfacing whether pay was justified by company performance, not just peer comparison
- Built a composite risk metric grounded in the Expected Loss framework to normalise across industries, preventing the misleading conclusion that larger sectors are structurally worse
- Revealed that high anomaly count doesn't equal high structural risk, showing that raw frequency alone misrepresents where governance risk concentrates

Stripe Merchant Support Agent

2025

Python, LangChain, ChromaDB, Sentence Transformers, Streamlit, LangSmith

[GitHub](#)

- Built and deployed production-grade conversational support agent on a scraped Stripe documentation knowledge base, with LLM-based intent classification routing queries across 6 action handlers including RAG Q&A, multi-turn flows, and guardrail enforcement
- Implemented session persistence across page reloads and server restarts, and an evaluation framework across 15 adversarial test cases scored by an LLM-as-judge

COMPETITIONS

- Rotman Design Challenge** Finalist, Top 7 of 43 Teams

March 2026

 - Designed Compass, an LLM-powered agent embedded in Manulife's app to help users navigate insurance coverage; validated through 7 empathy interviews and 38 survey responses which revealed that 73% of users left health benefits unused due to navigability failure, not product gaps.
- MMA Datathon** 1st Place

Jan 2026

 - Placed 1st across all competing teams; analysed 6,865 hours of web traffic data to evaluate paid search effectiveness for a retail brand operating across Canada and the US.
 - Applied causal inference to isolate true campaign lift from organic traffic; quantified 35% positive ROI in US markets, identified loss-making spend in Canada, and delivered a \$75K to \$90K budget reallocation recommendation to a judging panel under competition constraints.
- Koru Problem Hunt**

Nov 2025

 - Designed ThirdPlace, a dual-sided hobby marketplace; conducted mixed-methods research across hobbyists and studio owners, quantified three dominant barriers to participation, sized the North American hobby market at TAM of 5B, and built a full hypothesis register across desirability, feasibility, and viability.

EDUCATION

- University of Toronto, Rotman School of Management**
Master of Management Analytics (MMA) | Toronto, ON

2025 – 2026

Relevant coursework: Machine Learning, Statistical Modelling, Applications of LLMs, Analytical Insights Using Financial and Accounting Data
- IIIT-DM Kancheepuram**
B.Tech in Engineering, Minor in Product Design | Chennai, India

2021 – 2025

SKILLS

Languages: Python, SQL

ML and DL: scikit-learn, PyTorch, XGBoost, LangChain, sentence-transformers, Keras, Hugging Face

AI and NLP: RAG pipelines, LLM integration, vector search, text classification, intent routing

Explainability: DiCE counterfactuals, SHAP, Platt Scaling, fairness auditing, causal inference

Data and Infra: ChromaDB, SQLite, Streamlit, Flask, GitHub, Databricks, AWS, Tableau, Docker, Evidently

Domain: AML and regulatory compliance, credit risk, enterprise risk analytics, product analytics